

Lab 9: Structural Bioinformatics

AUTHOR

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Exploring the PDB Database

The main database for structural biology is PDB.

```
pdb <- read.csv("Data Export Summary.csv")
pdb
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	176,204	20,299	12,708	342	218
2	Protein/Oligosaccharide	10,279	3,385	34	8	11
3	Protein/NA	9,007	5,897	287	24	7
4	Nucleic acid (only)	3,066	200	1,553	2	15
5	Other	173	13	33	3	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron Other Total					
1	83 32	209,886				
2	1 0	13,718				
3	0 0	15,222				
4	3 1	4,840				
5	0 0	222				
6	0 4	22				

```
library(readr)
```

```
pdb <- read_csv("Data Export Summary.csv")
```



Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
total <- sum(pdb$Total)
xr <- sum(pdb$`X-ray`)
em <- sum(pdb$EM)

xrpercent <- (xr/total) *100
empercent <- (em/total) *100
```

The percentage of structures in the PDB solved by X-Ray is 81.48% and Electron Microscopy is 12.22%.

Q2: What proportion of structures in the PDB are protein?

```
prot <- as.integer(pdb[1,"Total"])
```

```
protpercent <- (prot/total) *100
```

The percentage of structures in the PDB that are proteins is 86.05%

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are 4865 HIV-1 Protease structures in the current PDB.

Exploring PDB Structures

```
library(bio3d)  
  
hiv <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
hiv
```

Call: read.pdb(file = "1hsg")

Total Models#: 1

Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)

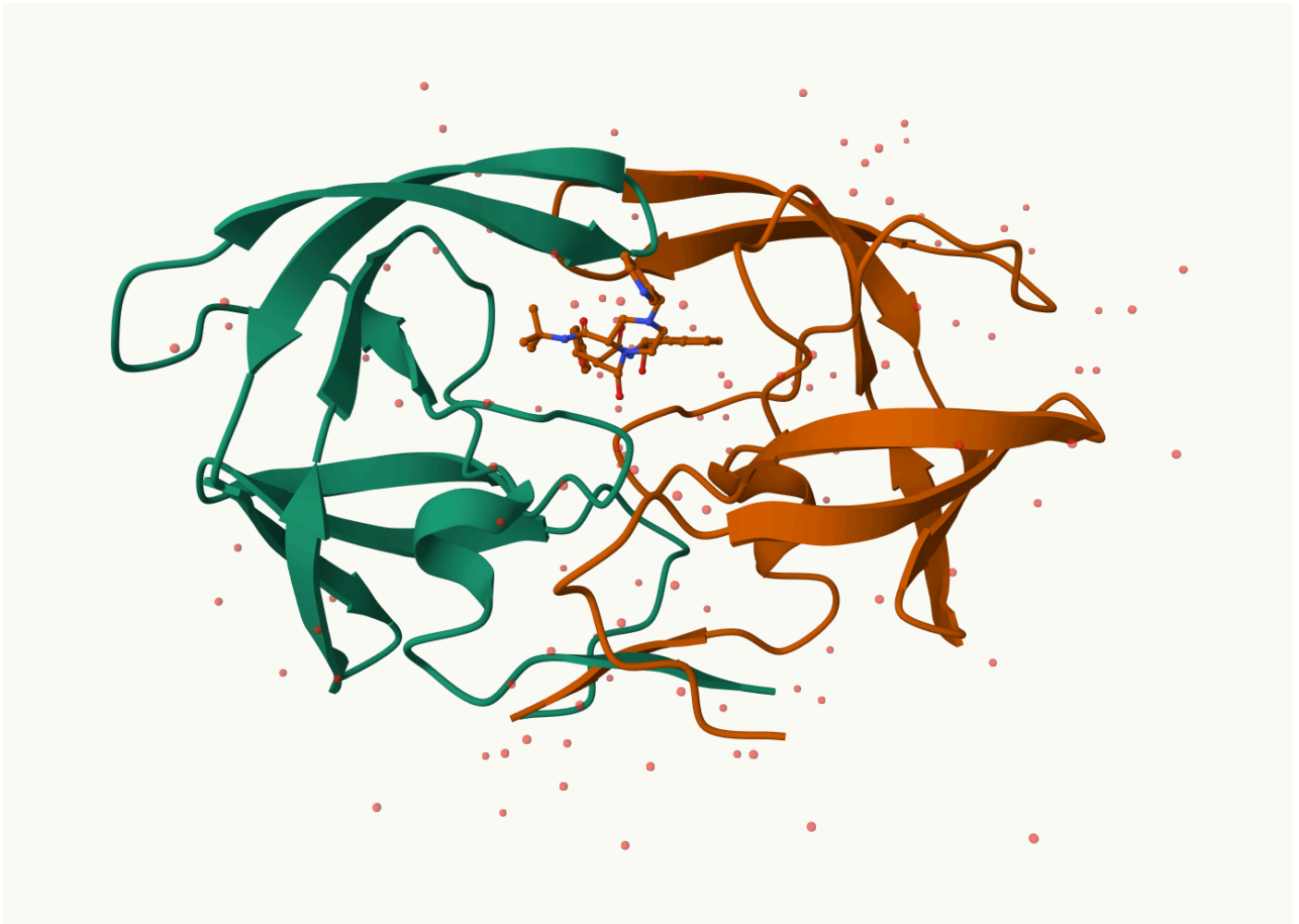
Non-protein/nucleic resid values: [HOH (127), MK1 (1)]

Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD  
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE  
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP  
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

Using Mol* viewer to explore the structure



My first view of HIV-Pr

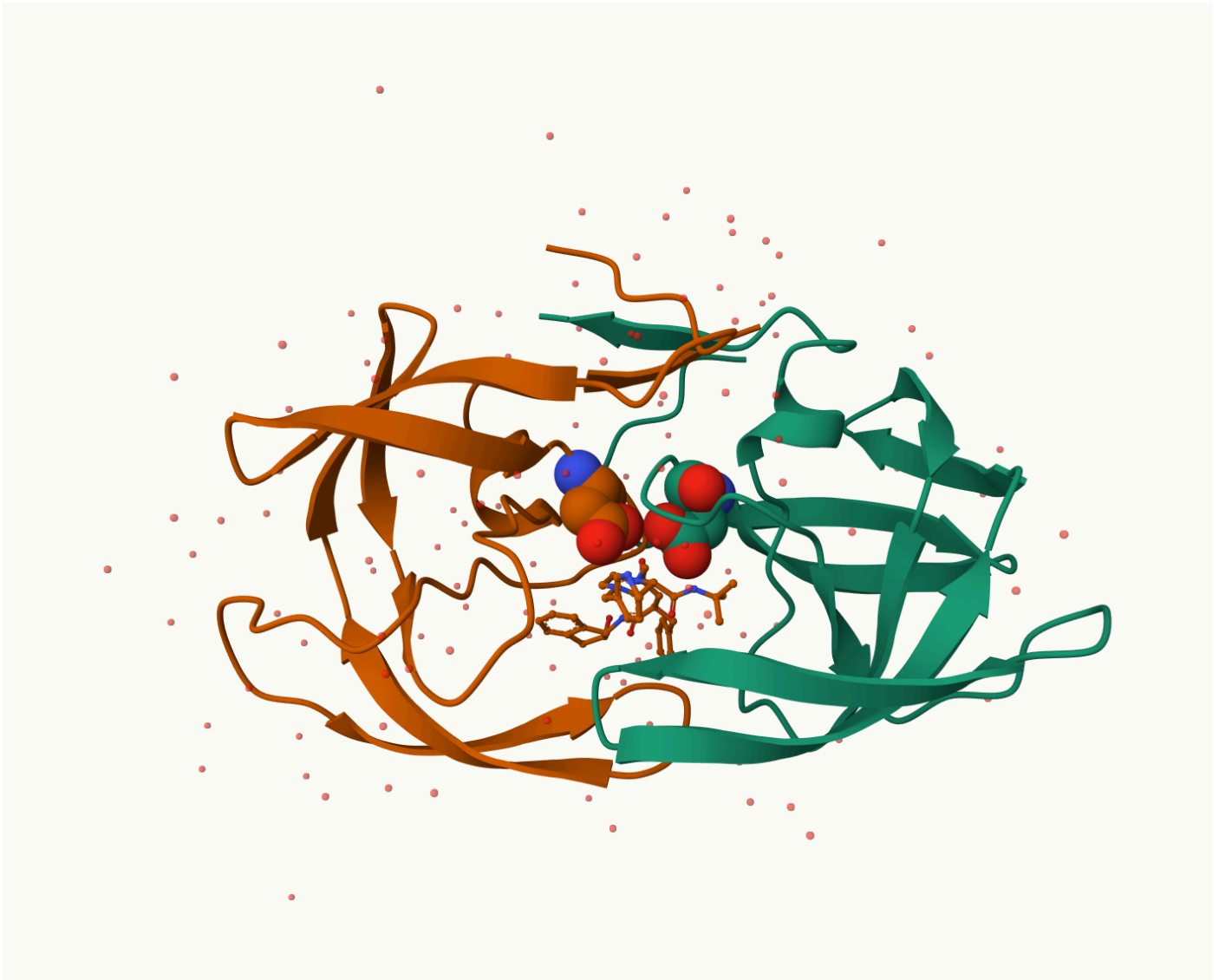
Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

This is because the Mol* viewer does not show hydrogen atoms.

Q5: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

The water molecule is residue number 304

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend "Ball & Stick" for these side-chains). Add this figure to your Quarto document.



ASP residues

hiv

```
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```

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Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
```

VNIIGRNLLTQIGCTLNF

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

198

Q8: Name one of the two non-protein residues?

HOH

Q9: How many protein chains are in this structure?

2

```
pdbseq(hiv)
```

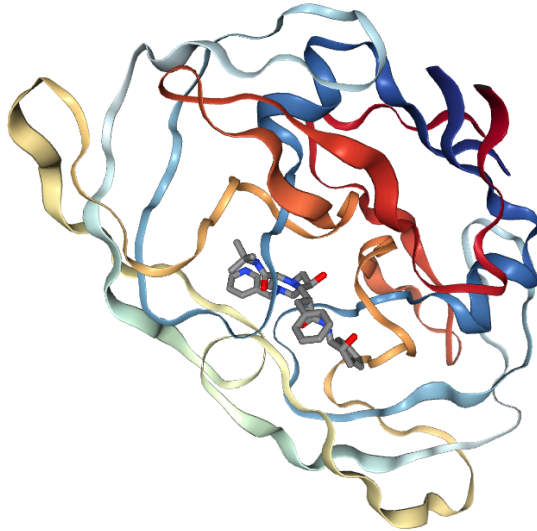
```
 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20
"P" "Q" "I" "T" "L" "W" "Q" "R" "P" "L" "V" "T" "I" "K" "I" "G" "G" "Q" "L" "K"
21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
"E" "A" "L" "L" "D" "T" "G" "A" "D" "D" "T" "V" "L" "E" "E" "M" "S" "L" "P" "G"
41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
"R" "W" "K" "P" "K" "M" "I" "G" "G" "I" "G" "G" "F" "I" "K" "V" "R" "Q" "Y" "D"
61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80
"Q" "I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T"
81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99  1
"P" "V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F" "P"
 2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21
"Q" "I" "T" "L" "W" "Q" "R" "P" "L" "V" "T" "I" "K" "I" "G" "G" "Q" "L" "K" "E"
22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41
"A" "L" "L" "D" "T" "G" "A" "D" "D" "T" "V" "L" "E" "E" "M" "S" "L" "P" "G" "R"
42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61
"W" "K" "P" "K" "M" "I" "G" "G" "I" "G" "G" "F" "I" "K" "V" "R" "Q" "Y" "D" "Q"
62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81
"I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T" "P"
82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99
"V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F"
```

```
chainA_seq <- pdbseq(trim.pdb(hiv, chain="A"))
```

```
# library(NGLViewer)
# install.packages("bio3d")
# library(bio3d)
# install.packages("bio3dview")
# library(bio3dview)
# install.packages("pak")
```

```
# pak::pak("bioboot/bio3dview")
# install.packages("NGLViewer")

# view.pdb(hiv) |> setSpin()
```



```
# view.pdc(hiv, colorScheme="sse")
# view.pdc(hiv, colorScheme="chain", col=c("blue","orange"), backgroundColor="pink")
```

Predict protein flexibility

We can run a bioinformatics analysis to predict protein dynamics i.e. functional motions

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
  Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)
```

Non-protein/nucleic resid values: [CL (3), HOH (238), MG (2), NA (1)]

Protein sequence:

```

MRIILLGAPGAGKGTQAFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
TDELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG

```

```

+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call

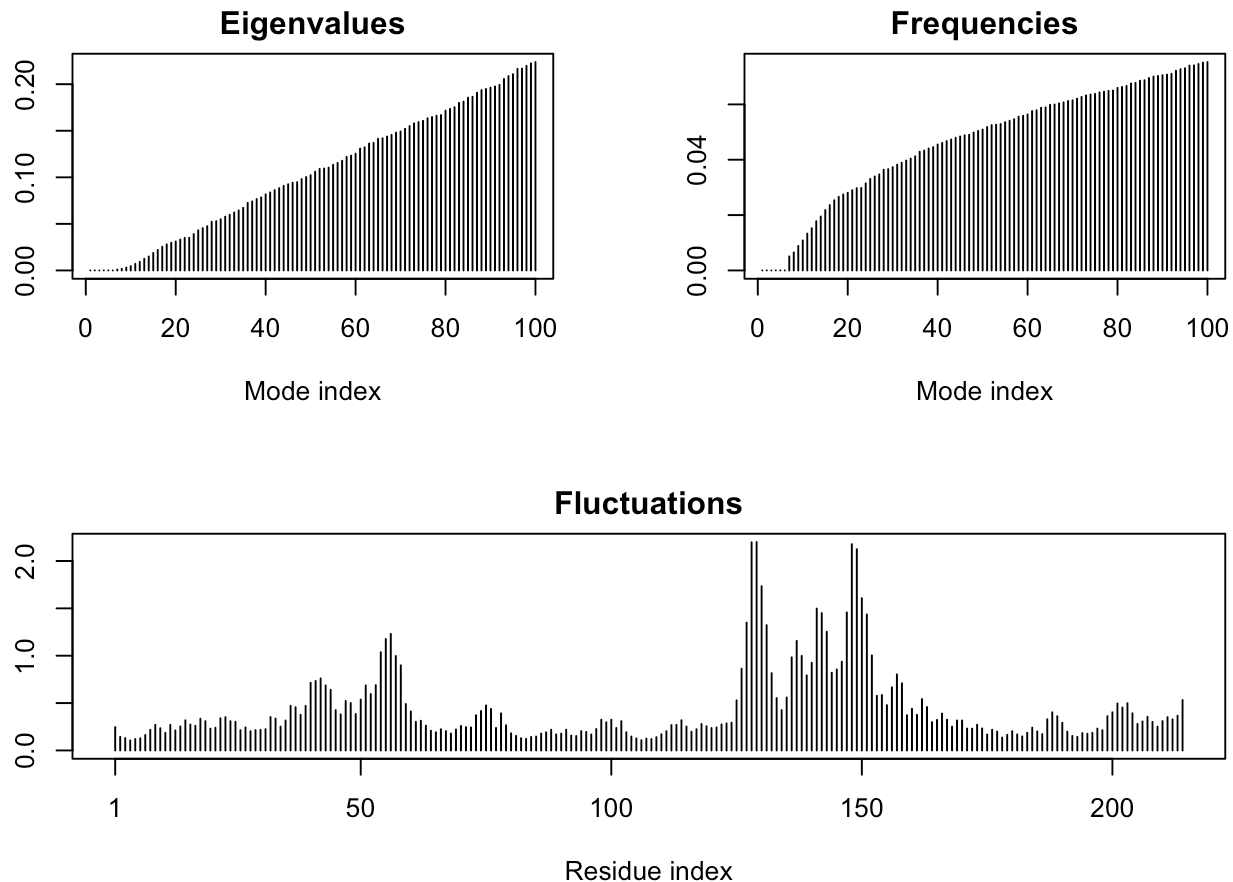
```

```
m <- nma(adk)
```

Building Hessian... Done in 0.014 seconds.

Diagonalizing Hessian... Done in 0.068 seconds.

```
plot(m)
```



Regions with peaks are predicted to be more flexible and dynamic than the flat regions.

Generate a "trajectory" of predicted motion

```
# mktrj(m, file="ADK_nma.pdb")
```



```
# bio3dview  
# view.nma(m, pdb=adk)
```